

Spearmint herbal tea has significant anti-androgen effects in polycystic ovarian syndrome. A randomized controlled trial.

Hirsutism in polycystic ovarian syndrome (PCOS), consequent to elevated androgen levels leads to significant cosmetic and psychological problems. Recent research in Turkey has shown that spearmint tea has antiandrogenic properties in females with hirsutism. No research has yet been undertaken to assess whether a reduction in androgen levels brought about by spearmint tea, translates to a clinical improvement in the degree of hirsutism. This study was a two centre, 30 day randomized controlled trial. Forty two volunteers were randomized to take spearmint tea twice a day for a 1 month period and compared with a placebo herbal tea. At 0, 15 and 30 days of the study serum androgen hormone levels and gonadotrophins were checked, the degree of hirsutism was clinically rated using the Ferriman-Galwey score and a questionnaire (the modified DQLI = Dermatology Quality of Life Index) was used to assess improvements in the level of self-reported hirsutism. Forty one of 42 patients completed the study. Free and total testosterone levels were significantly reduced over the 30 day period in the spearmint tea group ($p < 0.05$). LH and FSH also increased ($p < 0.05$). Patient's subjective assessments of their degree of hirsutism scored by the modified DQLI were significantly reduced in the spearmint tea group ($p < 0.05$). There was, however, no significant reduction in the objective Ferriman-Galwey ratings of hirsutism between the two trial groups over the trial duration ($p = 0.12$). There was a clear and significant alteration in the relevant hormone levels. This is associated clinically with a reduction in the self-reported degree of hirsutism but unfortunately not with the objectively rated score. It was demonstrated and confirmed that spearmint has antiandrogen properties, the simple fact that this does not clearly translate into clinical practice is due to the relationship between androgen hormones and follicular hair growth and cell turnover time. Simply put, the study duration was not long enough. The original studies from Turkey were in fact only 5 days long. The time taken for hirsutism to resolve is significant and a much longer future study is proposed as the preliminary findings are encouraging that spearmint has the potential for use as a helpful and natural treatment for hirsutism in PCOS. (c) 2009 John Wiley & Sons, Ltd.